DHP11021: Project Data Description and Management Plan
Assignment 1 - Template

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1. DATASET A

Dataset Description:

a. **Basic information**: Fill in the following table regarding your dataset. Add additional lines if your collection has multiple data files.

In/ Output #	Description of data source	Data type (e.g. text/images)	Data format (e.g., .txt; .csv)	File size (e.g. in MB/GB)	Origin/Access Conditions (license, public domain, etc.)
1	Roboflow handwriting detection: 361 annotated handwriting images sourced from the real-world handwritten documents. This dataset is accompanied by detection model.	Images	.jpg	55MB	License: CC BY 4.0
2					

b. **Data Collection**: Where and how was data collected (be specific) or produced?

The images were collected from a variety of real-world handwritten documents and digitized by photographing and scanning, and frame the handwritten content.

c. **Reason for Collection**: What do you find interesting about this dataset? What kinds of research questions do you think you might be able to ask and answer with it (either as the whole or part of your data)?

Interesting: This dataset was designed for the handwriting detection models, and it supports the research in document analysis, archives, and manuscript recognition. This dataset includes 361 images, and it is more suited for small-scale studies; it can also be a model to prove one's ideas.
Research questions and answers:
1. How much data is really needed to detect handwriting reliably?
---Extract samples of the 361 images, such as 50, 100, and 200 images, and compare the outcome to measure the efficiency of the data process.
2. How well do models trained on printed text transfer to handwritten text?
---According to the printed text dataset and make adjustments.
3. What is a better way to integrate handwritten detection with recognition?
---Using this dataset for detection, then put the handwritten images into a handwriting recognition system. Comparing their outcome.

Data Quality:

a. **Data Processing:** Will this data require and data cleaning or processing? If so those requirements and/or processes should be described here. How did you/would you document these changes, or at least make them reversible if required?

The images have been processed and they are turned the right way up. They are also edited in different ways, like rotating, stretching, and changing brightness. But this is a small-scale model; it can add more ancient manuscripts to better train and promote its function.

b. **Data Organisation**: How are you organising the data for storage? In one file or many? In one folder or many? Using what file-naming or other conventions are you using to aid findability?

The data is stored in many files; each image has its own annotation file. All files are kept in organized folders and I can use consecutive names so they can be easily found and processed.

c. **Data Structuring and Documentation**: Are there any structural elements, such as metadata or documentation (e.g., notes on the methodology of data collection, or way of organising data) that accompany the data? If you have structured data elements in your dataset yourself, where did this structure come from (was it determined by another tool, data creator, or did you choose the fields yourself such as when gathering or organising data)? Are any formalised standards being applied to the data in terms of how they will or have been organised (e.g., a metadata standard or controlled vocabulary)?

Each image in the dataset has boxes around the handwriting to show where it is in the picture. The dataset also includes extra details called metadata, like how the data was collected and what each part means. The creators explain the steps they used so that people can check the results. The images are split into three groups: training, validation, and testing, which helps make sure that the data used to build models is accurate.
By providing clear labels, instructions, and splits, this dataset is useful for people working on handwritten detection.

2. DATASET B

Dataset Description:

a. **Basic information**: Fill in the following table regarding your dataset. Add additional lines if your collection has multiple data files.

In/Output #	Description of data source	Data type (e.g. text/images)	Data format (e.g., .txt; .csv)	File size (e.g. in MB/GB)	Origin/Access Conditions (license, public domain, etc.)

1	The Community Arts, Culture, and Libraries dataset includes metadata on libraries, museums, and cultural facilities in South Dublin. Fields include name, address, contact information, facities (WiFi, parking, childcare, disability access, etc.), activities, ownership, and coordinates.	Structured data	.csv	3KB	Open data, public domain (South Dublin County Council)
2					

b. **Data Collection**: Where and how was data collected (be specific) or produced?

This dataset was collected and published by South Dublin County Council. It was probably edited from council records of libraries, museums, and cultural facilities. It also includes detailed metadata for each part.

c. **Reason for Collection**: What do you find interesting about this dataset? What kinds of research questions do you think you might be able to ask and answer with it (either as the whole or part of your data)?

This dataset provides an overview of community cultural infrastructure in South Dublin, which can help people to learn and preserve local culture. Additionally, this dataset is quite messy before it is processed.
Questions and answers:
1. What cultural facilities and resources are available to the public in South Dublin?
---This dataset lists 13 facilities.
(Libraries and archives: 7.
Museums: 2.
Theatre, drama, music centres: 2.
Community art/ drama/ craft centres: 1.
Cultural/ educational centres: 1.)
2. How accessible are these facilities? (for example, disability access, parking, wifi availability?)
---Disability access: 2 have access, 11 do not.
Parking: 2 have parking, 11 do not.
Wifi availability: 2 offer wifi, 11 do not.

Data Quality:

a. **Data Processing:** Will this data require and data cleaning or processing? If so those requirements and/or processes should be described here. How did you/would you document these changes, or at least make them reversible if required?

The dataset is relatively clean, but some fields still have missing or inconsistent values, and it needs a small clean.
Phone numbers appear in different formats; some include +353, others do not.
---Standardise to a single format, all with +353.
Parking/ Wifi/ DisabilityAccess columns may have inconsistent values (Yes, yes, No, no)
---Using Open Refine.
Use Facet—Text facet to review unique values.
Edit in the facet panel to unify all entries as Yes/No.

b. **Data Organisation**: How are you organising the data for storage? In one file or many? In one folder or many? Using what file-naming or other conventions are you using to aid findability?

The dataset is stored in a single .csv file. It can be stored in a folder named Community_Arts_Culture_Libraries_Research.csv. Maintaining backups and original versions for research.

c. **Data Structuring and Documentation**: Are there any structural elements, such as metadata or documentation (e.g., notes on the methodology of data collection, or way of organising data) that accompany the data? If you have structured data elements in your dataset yourself, where did this structure come from (was it determined by another tool, data creator, or did you choose the fields yourself such as when gathering or organising data)? Are any formalised standards being applied to the data in terms of how they will or have been organised (e.g., a metadata standard or controlled vocabulary)?

This dataset is already structured in Excel form, with clear column names and metadata. Metadata includes X, Y, and facility information, and details. It follows a typical open data structure; researchers can enrich it with other metadata standards when making different projects.

Data Management Plan

The following has been adapted from the Science Europe Core Requirements for Data Management Plans.

Data Preservation: Storage and backup during the research process

a. **Data Storage**: How/where will data and metadata be stored? Be very careful answering this question if you are using an external service requiring remote cloud storage, you may have to dig in a bit to their structure to understand where your data is (e.g., Google or

Amazon may store your data offshore, in potential breach of ethics or GDPR requirements).

All data and metadata will be securely stored on encrypted computers or servers, organized with clear folder structures and consistent file names. If cloud services are used, only GDPR-compliant providers will be chosen. Data will be saved in standard formats, such as CSV, JSON, with accompanying metadata.

b. **Data Security**: How will data be backed up during the research process (where, how often)? Do you face any specific issues related to protection of personal and sensitive data (see below) during the research?

All research and helpful data will be regularly backed up to secure servers and, if necessary, backed up the data into cloud storage. Data will be backed up once every three days, and some very important data will be backed up daily.
I think protecting personal data is necessary when we are doing research or writing academic papers. We need to keep the original ideas, but the reference list also needs to be updated; we must make annotations if we get some ideas from other research.

Legal and ethical requirements

a. **Personal and Sensitive Data**: Does your data set contain any *personal* data? If so, does it contain any *sensitive* data? If personal data are being processed, how will compliance with legislation on personal data and on data security be ensured (in particular GDPR)?

Dataset A and Dataset B do not contain any personal or sensitive data. All images do not contain information about individuals and therefore no GDPR or related compliance issues apply.

b. **Copyright and IPR**: How will other legal issues, such as intellectual property rights and ownership, be managed? What legislation is applicable?

These datasets use images and annotations that are openly licensed or created by the research team. Intellectual property rights are respected by following the license terms of the original sources. Ownership of the dataset is clearly attributed, and it is shared under an open license for research use. Applicable legislation includes copyright law and open data licensing frameworks.

c. **Other ethical issues**: How will possible ethical issues be taken into account, and codes of conduct followed? Was there any ethical oversight sought or offered in the project (was this considered?)?

The project includes low ethical risk because it does not involve personal or sensitive data. Ethical issues were considered during planning, and openly licensed images or data were used to avoid misuse. These projects follow standard research codes of conduct, including proper citation and respect for data license. Formal ethical oversight was not required due to the non-sensitive data.

Data sharing and long-term preservation

a. **Open Data**. Although you may not be able to foresee this now, is possible you will want to share you data openly? How and when? Are there possible restrictions to data sharing or embargo reasons?

Yes, it is possible that I will want to share my data openly when I finalize and have already checked the quality of my project. It will be made available through platforms such as GitHub. There are no restrictions or embargoes expected, since the dataset does not contain any personal or sensitive data and only uses openly licensed or public domain materials.

b. **Data Disposal**. How long will you maintain your data? After that point, how will it be de-commissioned? (note: this question may provide an opportunity to think about how you keep data on your laptop generally, such as when you delete old files or if you may lose items when switching machines. Can your overall data management processes be improved, or at least be made more explicit?)

The data will be maintained for the duration of the project and stored securely on backed-up cloud storage. After this period, if the data is no longer needed, the local copies will be securely deleted, and the final version will be stored in trusted repositories, such as GitHub.
In general, files on personal devices are often backed up, and old files are deleted to improve data management and avoid accidental loss when changing machines.

c. **Formats and Acce**ss. What methods or software tools will be needed to access and use the data? Do these pose any sustainability risks or limitations to reuse?

The data can be accessed using standard tools such as Python, Open Refine, Excel, and Zotero.
There are actually some sustainability risks, for example, data stored in Excel may become unreadable, some software or website may break because of time and a lack of maintenance. To reduce these risks, we should prefer well-documented formats (CSV, JSON, etc.)

Data management responsibilities and resources

a. **Data Stewardship.** Who (for example role, position, and institution) will be responsible for data management (i.e. the data steward)? Note that legally, your data controller is likely TCD, but you are the data processor. These are slightly different from being the data steward, who looks after the preservation and continuity of the data.

The data steward will be the relevant staff in TCD library or IT Services, responsible for data preservation and management, and I (the researcher) am the data processor, and TCD is the data controller.

b. **FAIR Data.** What resources (for example financial and time) will be dedicated to data management and ensuring that data will be FAIR (Findable, Accessible, Interoperable, Re-usable)? It may be useful to remember that making data FAIR is not the same as making data open. The first (and sometimes only) user a FAIR approach may benefit may be yourself. You may find some of the enablers of FAIR are discussed elsewhere in the document (stability of technology choices, use of standards, rights status of data, storage and file naming conventions, etc.) if so, you need only to briefly reference them again here for their role as FAIR enablers.

Time will be dedicated to data management, creating metadata, using platforms to clean the messy data, and standard naming conventions. There may be no dedicated financial costs beyond existing institutional resources if some institution provides storge and repository services.

Reference list

Roboflow. 2022. “Handwriting Detection Dataset.” Accessed October 2, 2025. https://universe.roboflow.com/handwriting-detection-zfowd/handwriting-detection-eoxpj

South Dublin County Council. 2021. “Arts, Culture, and Libraries Dataset.” Updated April 13, 2021. https://data.gov.ie

This template utilized ChatGPT-5 (OpenAI, https://chatgpt.com/) to search for specialized information about the data management plan. All analyses of dataset A and dataset B are my own.